

# MUHAMMED RAIHANUL HOQUE

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📍 Paderborn, Germany



## PROFILE

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Master's student in Electrical Systems Engineering with hands-on experience in high-speed analog front-end circuit design for optical communications. Developed a fully differential transimpedance amplifier (TIA) for a coherent IQ receiver as part of my thesis with full schematic-to-layout flow including DRC, LVS, PEX, and EM simulation. Proficient in Cadence Virtuoso and in MATLAB. Focused on building a deep understanding of AMS IC design through project work, and looking to continue that in an industry environment across Europe.

## EDUCATION

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**M.S. in Electrical Systems Engineering** 2021-2026(Within Sep)  
Paderborn University, NRW, Germany German Grade: 2.0

- **Master's Thesis:** Design of a Low-Noise Fully Differential Transimpedance Amplifier (TIA) with DC Offset Compensation for Coherent IQ Receiver (Grade: 1.3).
- **Key Results:** Achieved 80 GHz bandwidth (160 Gb/s NRZ) and 14 pA/ $\sqrt{\text{Hz}}$  input-referred noise, 37 dB dynamic range, and 5mA DC offset compensation using IHP 130 nm SiGe BiCMOS.
- **Workflow:** Topology definitions, schematic design, corner simulations, layout design (shielding, matching, symmetry, guard ring), DRC and LVS verifications, PEX extraction, EM simulation (RFPro).
- **Coursework:** Circuit and System Design, IC Design for Wireline Communications, RF IC Design, Fields and Waves, Advanced Systems Theory.

**B.S. in Electrical and Electronic Engineering** 2013-2017  
Ahsanullah University of Science and Technology, Dhaka, Bangladesh German Grade: 2.3

- **Coursework:** Electronics I & II, Solid State Devices, Electromagnetics, Digital Electronics, Circuit Theory, Control System.

## PROJECTS

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**Two-Stage Miller Compensated OpAmp Design using  $g_m/I_D$  methodology:** 09.22-11.22

- Key Specs: DC Gain: 52 dB, Phase Margin: 67°, Unity Gain Frequency (UGF): 60 MHz, PSRR :60/70 dB (Pos/Neg), CMRR: 60 dB, DC Power: 550  $\mu\text{W}$ .

**RF/Optical Receiver Design (IHPs 130nm BiCMOS G3)** 04.23-09.23

- Design of a differential TIA and comparison between feedforward and shunt feedback TIA in terms of noise and BW performance.

**Dielectric Slab Waveguide Mode Solver** 04.24-09.24

- Designed a quasianalytic mode solver for anisotropic thin film lithium niobate (TFLN) slabs.

## WORK EXPERIENCE

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**Graduate Research Assistant** 11/2023 – 05/2024  
Heinz-Nixdorf Institute, Paderborn University, Germany

- Contributed to build a system architecture for a digital-twin of Paderborn-Lippstadt airport (passanger boarding pathway) using SysML language and Cameo Systems Modeler.

**Assistant Engineer** 12/2023 – 04/2021  
KSRM Billet Industries, Chittagong, Bangladesh.

- Developed, tested and optimized PLC (programmable logic controller) programs for controlling industrial processes and machineries with team.

## TECHNICAL SKILLS

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**Circuit Knowledge:** Differential Amplifiers, Current Mirrors, Biasing Circuits, Cascodes, OpAmps, CTLE, Driver, Oscillators, MZM.  
**EDA Tools:** Cadence Virtuoso (Layout XL, ADE, Spectre), RFPro, LTspice.  
**Physical Verification:** Assura/PVS DRC, LVS, PEX extraction.  
**Scripting/OS:** SKILL, Python (Basic), MATLAB, Linux.

## LANGUAGES

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**Bengali:** Native    **English:** C1    **German:** Basic